

- a) Denoting with $Y_i = \text{mpg}_i$ $i = 1, \dots, 32$ (response variable)
 and with $x_{ij} = \text{value of the } j\text{-th covariate on the } i\text{-th car}$
 $j = 1, \dots, 9$ $i = 1, \dots, 32$
 (with $x_{i0} = 1$ for all i , since the model includes the intercept)

The model can be written as

$$Y_i = \beta_0 + \beta_2 \underbrace{x_{i2}}_{wt_i} + \beta_3 \underbrace{x_{i3}}_{am_i} + \dots + \beta_9 \underbrace{x_{i9}}_{vs_i} + \varepsilon_i \quad (\text{model A})$$

with $\varepsilon_i \stackrel{iid}{\sim} N(0, \sigma^2)$, $i = 1, \dots, 32$.

- b) Sample space $\mathcal{Y} = \mathbb{R}^{32}$
 Parameter space $\Theta = \mathbb{R}^9 \times (0, +\infty)$

- c) 1. t-value of "hp"

this value corresponds to the observed test statistic for testing the hypothesis

$$\begin{cases} H_0: \beta_6 = 0 \\ H_1: \beta_6 \neq 0 \end{cases}$$

$$t^{obs} = \frac{\hat{\beta}_6 - 0}{\sqrt{\hat{\text{var}}(\hat{\beta}_6)}} = \frac{\hat{\beta}_6}{\hat{\text{se}}(\hat{\beta}_6)} = \frac{-0.0214}{0.0162} = -1.321$$

2. estimate of "qsec"

this value is $\hat{\beta}_{qsec} = \hat{\beta}_8$ (max. lik. estimate)

we can derive it inverting the formula used in point 1.

$$t^{obs} = \frac{\hat{\beta}_8}{\hat{\text{se}}(\hat{\beta}_8)} \Rightarrow \hat{\beta}_8 = t^{obs} \cdot \hat{\text{se}}(\hat{\beta}_8) = 1.229 \cdot 0.6587 = 0.8095$$

3. $\Pr(>|t|)$ of "vs"

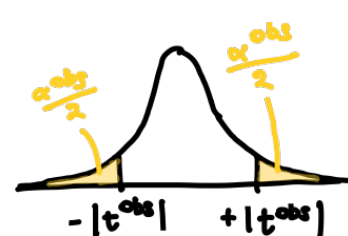
this is the pvalue of the test of significance of $\beta_9 = \beta_3$

$$\begin{cases} H_0: \beta_9 = 0 \\ H_1: \beta_9 \neq 0 \end{cases}$$

the test statistic (pivotal quantity) is

$$T = \frac{\hat{\beta}_9}{\sqrt{\hat{\text{var}}(\hat{\beta}_9)}} \stackrel{H_0}{\sim} t_{n-p} = t_{32-9} = t_{23}$$

$$\begin{aligned} \text{the pvalue is } \alpha^{obs} &= \Pr_{H_0}(|T| > |t^{obs}|) \\ &= 2 \Pr_{H_0}(T > |t^{obs}|) \\ &= 2 \Pr_{H_0}(T > 0.183) \end{aligned}$$



where $T \sim t_{23}$
 Student's t with 23 dof.

- d) We want to test

$$\begin{cases} H_0: \beta_2 = \beta_3 = \dots = \beta_8 = \beta_9 = 0 \\ H_1: \text{at least one of } \beta_j \text{ is } \neq 0 \quad (j=2, \dots, 9) \end{cases}$$

we use the statistic

$$F = \frac{\hat{\sigma}^2 - \hat{\sigma}^2}{\hat{\sigma}^2} \cdot \frac{n-p}{p-1} \stackrel{H_0}{\sim} F_{p-1, n-p}$$

with $\hat{\sigma}^2$ estimate of σ^2 under the null model $\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (y_i - \bar{y})^2 = \frac{1}{n} SST$ (H_0)

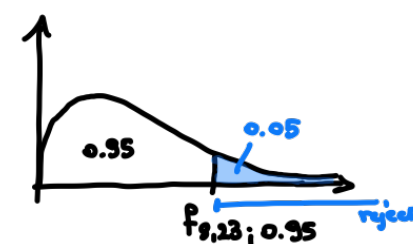
$\hat{\sigma}^2$ estimate under the full model "A" $\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \frac{1}{n} SSE$ (H_1)

$$F = \frac{SST - SSE}{SSE} \cdot \frac{23}{8} = \frac{SSR/SST}{SSE/SST} \cdot \frac{23}{8} = \frac{R^2}{1-R^2} \cdot \frac{23}{8}$$

$$\text{with the data I get } f^{obs} = \frac{0.9678}{1-0.9678} \cdot \frac{23}{8} = 18.87$$

at a significance level of 5%: I reject H_0 if $f^{obs} > \underset{2.3748}{f_{8,23;0.95}}$

hence I reject H_0 .



- e) model B

$$Y_i = \beta_1 + \beta_2 \underbrace{x_{i2}}_{wt_i} + \beta_3 \underbrace{x_{i3}}_{am_i} + \varepsilon_i$$

We can use a Test for comparing nested models, since model B can be obtained as a restriction of model A. Specifically, we need to test

$$\begin{cases} H_0: \beta_4 = \beta_5 = \dots = \beta_8 = \beta_9 = 0 \\ H_1: \text{at least one of } \beta_j \text{ is } \neq 0 \quad (j=4, \dots, 9) \end{cases}$$

the test statistic is

$$F = \frac{\hat{\sigma}^2 - \hat{\sigma}^2}{\hat{\sigma}^2} \cdot \frac{n-p}{p-p_0} \stackrel{H_0}{\sim} F_{p-p_0, n-p}$$

where

$\hat{\sigma}^2$ estimate of σ^2 under model B (H_0) $\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_{i,B})^2 = \frac{SSE_B}{n}$

$\hat{\sigma}^2$ estimate under the full model A $\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_{i,A})^2 = \frac{SSE_A}{n}$

$n = 32$

$p = 9$ $p_0 = 3$

$$SSE_A = (n-p) \cdot (\text{residual SE})^2 = 23 \cdot 2.544^2 = 148.85$$

$$SSE_B = (n-p_0) \cdot (\text{residual SE})^2 = 29 \cdot 3.098^2 = 278.33$$

$$f^{obs} = \frac{278.33 - 148.85}{148.85} \cdot \frac{23}{6} = 3.334$$

using a 5% significance level, I reject H_0 if $f^{obs} > \underset{2.527}{f_{6,23;0.95}}$

Hence, I reject H_0 : I prefer model A.

using a 1% significance level, I reject H_0 if $f^{obs} > \underset{3.71}{f_{6,23;0.99}}$

Here I do not reject H_0

- f) introducing the interaction we obtain the model

$$Y_i = \beta_1 + \beta_2 x_{i2} + \beta_3 x_{i3} + \beta_4 x_{i4} + \varepsilon_i$$

where $x_{i2} = wt_i$ (weight of car i)

$$x_{i3} = am_i = \begin{cases} 1 & \text{if car } i \text{ has manual transmission} \\ 0 & \text{if car } i \text{ has automatic transmission} \end{cases}$$

$$x_{i4} = x_{i2} \cdot x_{i3} = \begin{cases} x_{i2} = wt_i & \text{if } am_i = 1 \quad (\text{weight of car } i \text{ if manual}) \\ 0 & \text{if } am_i = 0 \end{cases}$$

interpretation of parameters

- consider a car with automatic transmission ($x_{i3} = 0$, $x_{i4} = 0$)

the mean consumption is

$$\mu_i = \beta_1 + \beta_2 x_{i2}$$

- for a car with manual transmission ($x_{i3} = 1$, $x_{i4} = x_{i2}$)

$$\begin{aligned} \mu_i &= \beta_1 + \beta_2 x_{i2} + \beta_3 + \beta_4 x_{i2} \\ &= (\beta_1 + \beta_3) + (\beta_2 + \beta_4) x_{i2} \end{aligned}$$

β_1 is the intercept for cars with an automatic transmission

$\beta_1 + \beta_3$ is the intercept for cars with a manual transmission

β_2 is the effect on the mean consumption of increasing the weight of the car of 1 unit, for cars with an automatic transmission

$\beta_2 + \beta_4$ is the effect on the mean consumption of increasing the weight of the car of 1 unit, for cars with a manual transmission

EXERCISE 2

$$n = 30 \quad (y_1, \dots, y_{10}, y_{11}, \dots, y_{30})$$

$$x_i = \begin{cases} 1 & i = 1, \dots, 10 \\ 0 & i = 11, \dots, 30 \end{cases}$$

$$Y_i \sim \text{Ber}(\pi_i) \quad \text{logit}(\pi_i) = \beta_1 + \beta_2 x_i$$

$$\Leftrightarrow \log \frac{\pi_i}{1-\pi_i} = \beta_1 + \beta_2 x_i \quad \Leftrightarrow \pi_i = \frac{e^{\beta_1 + \beta_2 x_i}}{1 + e^{\beta_1 + \beta_2 x_i}}$$

$$a) \quad P(y_i; \pi_i) = \pi_i^{y_i} (1-\pi_i)^{1-y_i}$$

$$P(y_1, \dots, y_{30}; \underline{\pi}) = \prod_{i=1}^{30} \pi_i^{y_i} (1-\pi_i)^{1-y_i}$$

likelihood function

$$L(\underline{\pi}) = \prod_{i=1}^{30} \pi_i^{y_i} (1-\pi_i)^{1-y_i} \rightarrow L(\underline{\beta}) = \prod_{i=1}^{30} \left(\frac{e^{\beta_1 + \beta_2 x_i}}{1 + e^{\beta_1 + \beta_2 x_i}} \right)^{y_i} \left(\frac{1}{1 + e^{\beta_1 + \beta_2 x_i}} \right)^{1-y_i}$$

log-likelihood function

$$\ell(\underline{\pi}) = \log L(\underline{\pi})$$

$$= \sum_{i=1}^{30} y_i \underbrace{\log \pi_i}_{\beta_1 + \beta_2 x_i} + (1-y_i) \underbrace{\log(1-\pi_i)}_{-\log(1 + e^{\beta_1 + \beta_2 x_i})}$$

$$= \sum_{i=1}^{30} y_i (\beta_1 + \beta_2 x_i) - \sum_{i=1}^{30} \log(1 + e^{\beta_1 + \beta_2 x_i})$$

hence

$$\ell(\underline{\beta}) = \sum_{i=1}^{30} y_i (\beta_1 + \beta_2 x_i) - \sum_{i=1}^{30} \log(1 + e^{\beta_1 + \beta_2 x_i})$$

$$= \sum_{i=1}^{30} \left\{ y_i (\beta_1 + \beta_2 x_i) - \log(1 + e^{\beta_1 + \beta_2 x_i}) \right\}$$

$$= \beta_1 \sum_{i=1}^{30} y_i + \beta_2 \sum_{i=1}^{30} x_i y_i - \sum_{i=1}^{30} \log(1 + e^{\beta_1 + \beta_2 x_i})$$

finally, the score function is

$$s_j(\underline{\beta}) = \frac{\partial \ell(\underline{\beta})}{\partial \beta_j} \quad j = 1, 2$$

$$= \begin{cases} \frac{\partial \ell(\underline{\beta})}{\partial \beta_1} = \sum_{i=1}^{30} y_i - \sum_{i=1}^{30} \frac{1}{1 + e^{\beta_1 + \beta_2 x_i}} \cdot e^{\beta_1 + \beta_2 x_i} \\ \frac{\partial \ell(\underline{\beta})}{\partial \beta_2} = \sum_{i=1}^{30} x_i y_i - \sum_{i=1}^{30} \frac{1}{1 + e^{\beta_1 + \beta_2 x_i}} \cdot e^{\beta_1 + \beta_2 x_i} \cdot x_i \end{cases}$$

The fitted model is

$$Y_i \sim \text{Ber}(\hat{\pi}_i) \quad \text{logit}(\hat{\pi}_i) = 1.3863 - 2.0794 x_i$$

$$b) \quad \hat{\pi}_i \text{ when } x_i = 0 \text{ is}$$

$$P(Y_i = 1 | x_i = 0) = \frac{e^{\hat{\beta}_1}}{1 + e^{\hat{\beta}_1}} = 0.800$$

$\hat{\pi}_i$ when $x_i = 1$ is

$$P(Y_i = 1 | x_i = 1) = \frac{e^{\hat{\beta}_1 + \hat{\beta}_2}}{1 + e^{\hat{\beta}_1 + \hat{\beta}_2}} = 0.333$$

when $x_i = 0$ the odds are

$$\frac{\text{prob. success} | x_i = 0}{\text{prob. failure} | x_i = 0} = \frac{P(Y_i = 1 | x_i = 0)}{P(Y_i = 0 | x_i = 0)} = \frac{\left(\frac{e^{\hat{\beta}_1}}{1 + e^{\hat{\beta}_1}} \right)}{\left(\frac{1}{1 + e^{\hat{\beta}_1}} \right)} = \frac{0.800}{0.200} = 4.00 \quad (= e^{\hat{\beta}_1})$$

odds $\cdot 100 = 400 =$ number of expected successes every 100 failures

$\rightarrow$ when $x = 0$, I expect 400 successes every 100 failures

when $x_i = 1$ the odds are

$$\frac{\text{prob. success} | x_i = 1}{\text{prob. failure} | x_i = 1} = \frac{P(Y_i = 1 | x_i = 1)}{P(Y_i = 0 | x_i = 1)} = \frac{\left(\frac{e^{\hat{\beta}_1 + \hat{\beta}_2}}{1 + e^{\hat{\beta}_1 + \hat{\beta}_2}} \right)}{\left(\frac{1}{1 + e^{\hat{\beta}_1 + \hat{\beta}_2}} \right)} = \frac{0.333}{0.666} = 0.500 \quad (= e^{\hat{\beta}_1 + \hat{\beta}_2})$$

$\rightarrow$ when $x = 1$, I expect 50 successes every 100 failures

Finally, the odds ratio is

$$\frac{\left(\frac{\pi_i}{1-\pi_i} \mid x_i = 1 \right)}{\left(\frac{\pi_i}{1-\pi_i} \mid x_i = 0 \right)} = \frac{e^{\hat{\beta}_1 + \hat{\beta}_2}}{e^{\hat{\beta}_1}} = e^{\hat{\beta}_2} = 0.1250$$

The odds for the group $x_i = 0$ are multiplied by 0.1250 to obtain the odds at $x_i = 1$

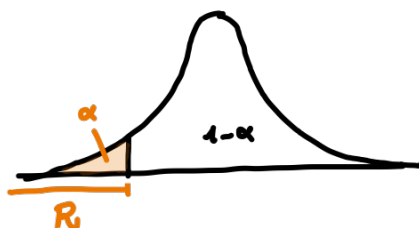
$$c) \quad \begin{cases} H_0: \beta_2 = -1 \\ H_1: \beta_2 < -1 \end{cases}$$

$$\text{The test statistic } Z = \frac{\hat{\beta}_2 - (-1)}{\sqrt{[j(\hat{\beta})]_{2,2}^{-1}}} \quad H_0 \sim N(0, 1)$$

from the summary

$$\sqrt{[j(\hat{\beta})]_{2,2}^{-1}} = 0.7926 \quad \hat{\beta}_2 = -2.0794$$

$$z_{\text{obs}} = \frac{-2.0794 + 1}{0.7926} = -1.3792$$



The reject region here is for negative values

Using a significance level α , I reject H_0 if $z_{\text{obs}} < -2\alpha$

$\alpha = 5\%$ $2\alpha = 2 \cdot 0.05 = -2 \cdot 0.95 = -1.64$ I do not reject H_0 at 5% level

$\alpha = 10\%$ $2\alpha = 2 \cdot 0.10 = -2 \cdot 0.90 = -1.28$ I reject H_0 at a 10% level

d) The residual deviance is the lik. ratio test between the saturated model and the proposed model:

$$D(\text{model}) = 2 \{ \tilde{\ell}(\text{saturated}) - \hat{\ell}(\text{model}) \}$$

where $\tilde{\ell}(\text{saturated})$ is the maximum of the log-likelihood under a model with n parameters,

and $\hat{\ell}(\text{model})$ is the maximum of the log-likelihood under the current model

The null deviance is

$$D(\text{null}) = 2 \{ \tilde{\ell}(\text{saturated}) - \tilde{\ell}(\text{null}) \}$$

where $\tilde{\ell}(\text{null})$ is the maximum of the log-likelihood under a model with a single parameter π